Physical Chemistry CUME, Sep. 8th, 2017

Thermo

Q1. State the first law of thermodynamics

Q2. State the second law of thermodynamics

Q3, State the third law of thermodynamics.

Q4. According to the Clausius theorem, the entropy ΔS is related to heat exchange δQ in a quasi-static near-equilibrium process through the equation, $\Delta S = \frac{\delta Q}{T}$. For most crystals, it has been shown that the heat capacity C_p is proportional to T^3 at sufficiently low temperature (assume the proportionality constant is *a, i.e.* $C_p = a \cdot T^3$). Please provide a schematic plot of entropy of a perfect crystal as a function from temperature from 0K to 2K. (2 K can be considered to be a sufficiently low temperature.) (Hint: What is the relationship between δQ and C_p ?, you may get partial credit if you provide a very good guess.)